

Lecture Notes: Graph Theory, Introduction to Hamiltonian Circuits and Paths

Math F113X Math and Society

Terminology

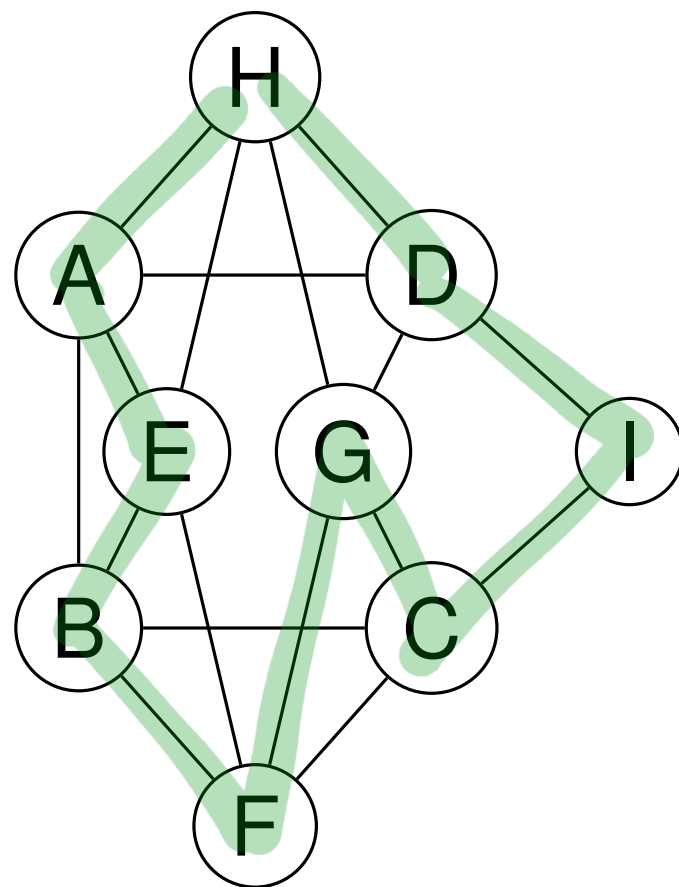
A **Hamiltonian circuit** (sometimes called Hamiltonian Cycle) is

A circuit that passes through every vertex exactly once (not every edge needs to be used)

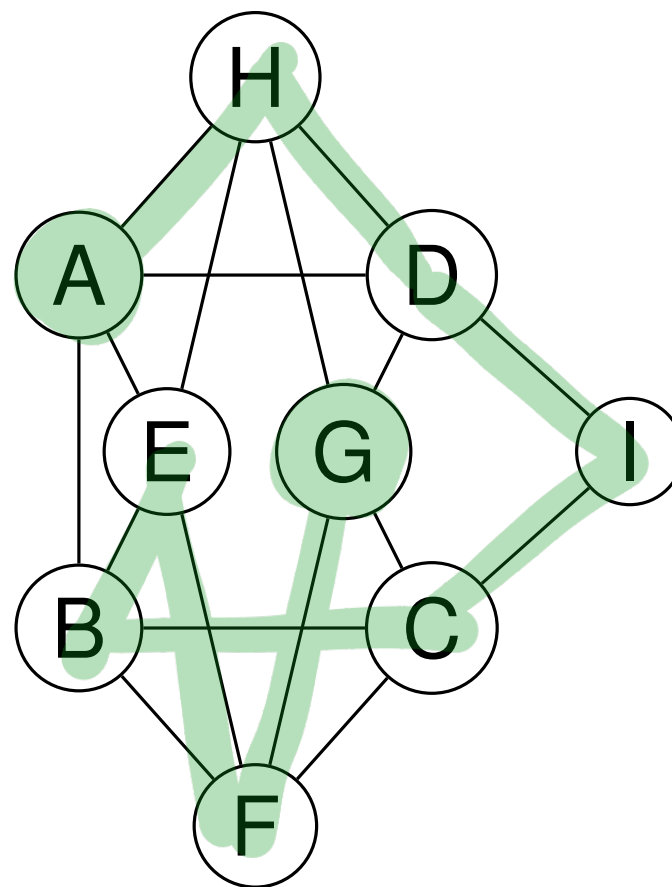
A **Hamiltonian path** is

A path that uses every vertex exactly once

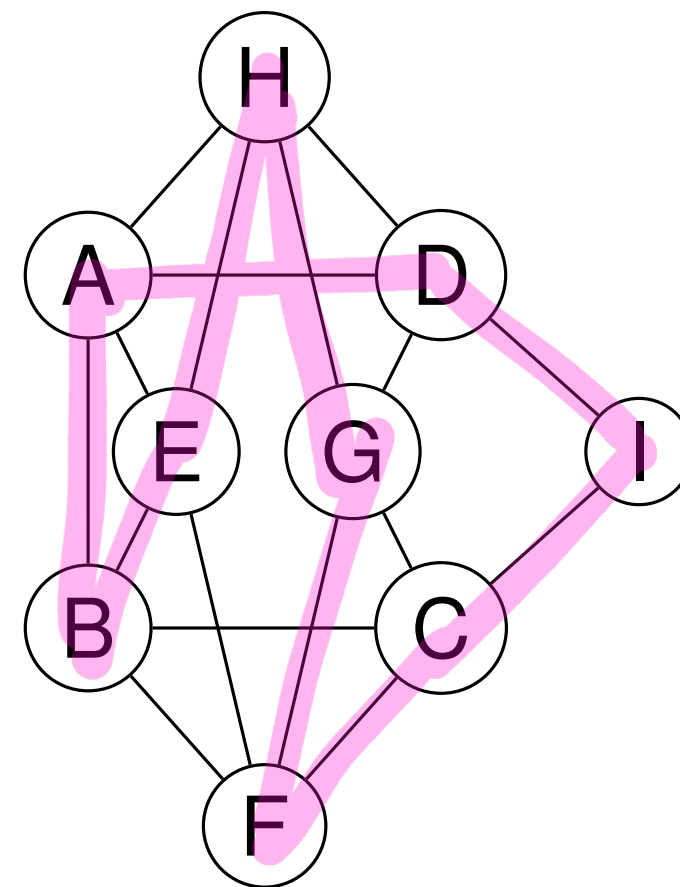
Some Example Graphs



Hamiltonian
Circuit



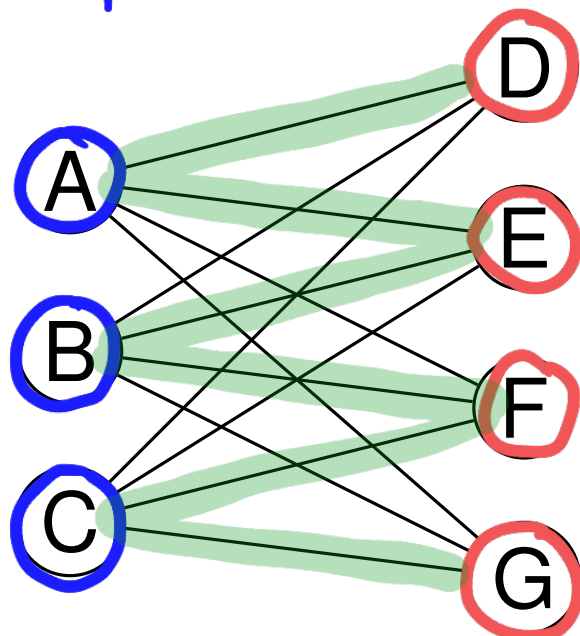
Hamiltonian path
(that's not a
circuit)



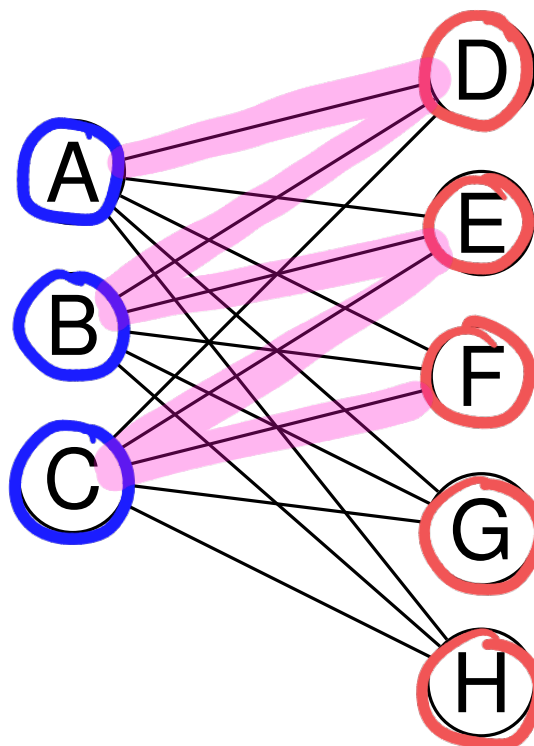
A different
Hamiltonian
Circuit

Some Example Graphs

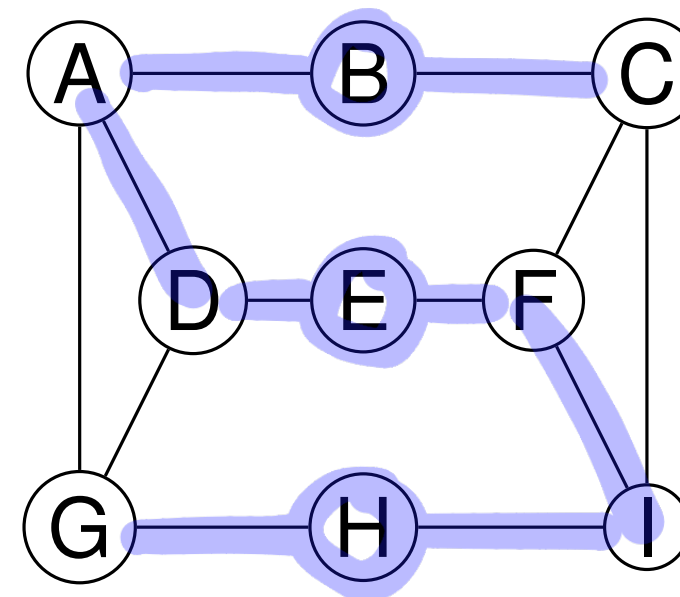
complete
(Bipartite Graph)



*Hamiltonian path,
no circuit is
possible*

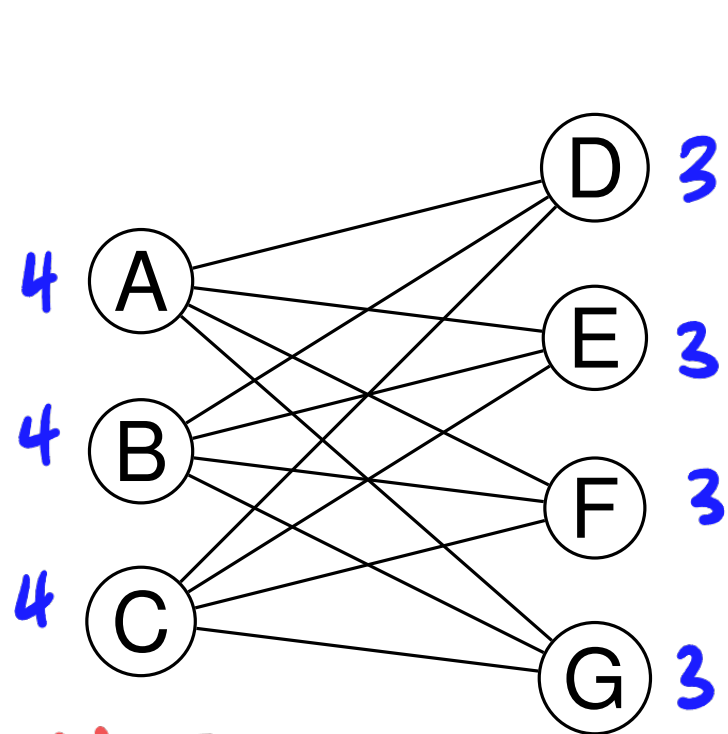


*Neither Hamiltonian
path nor Hamiltonian
Circuit!*

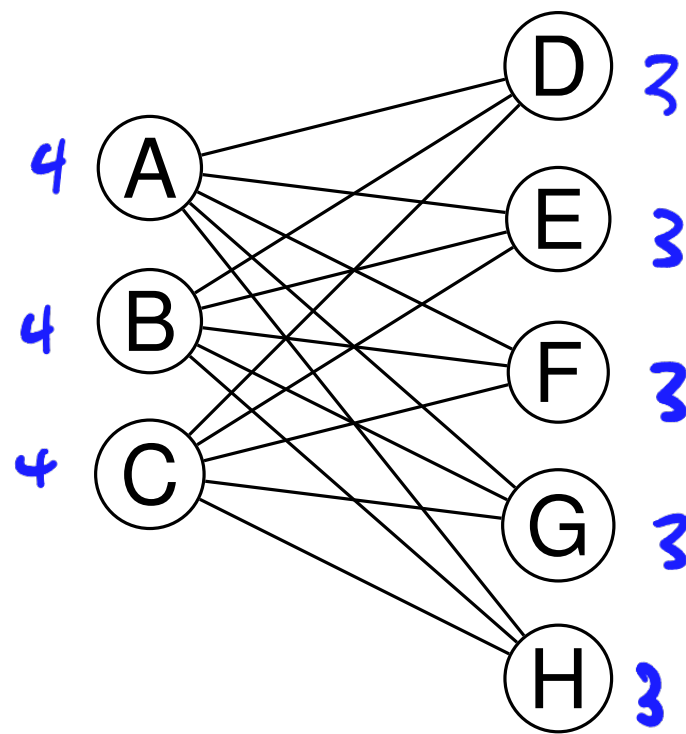


*Hamiltonian path,
but no circuit.*

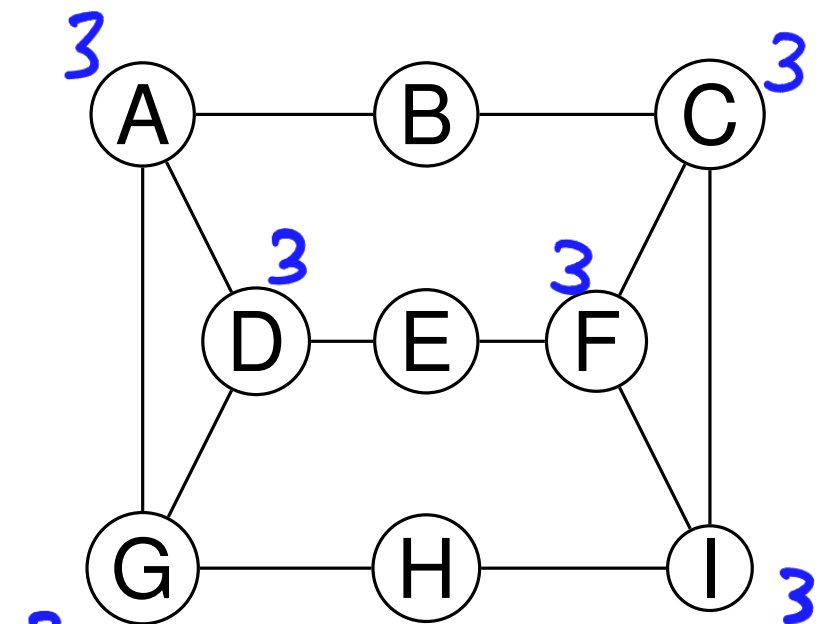
Do any of the graphs have an **Euler** circuit?



No EC or EP



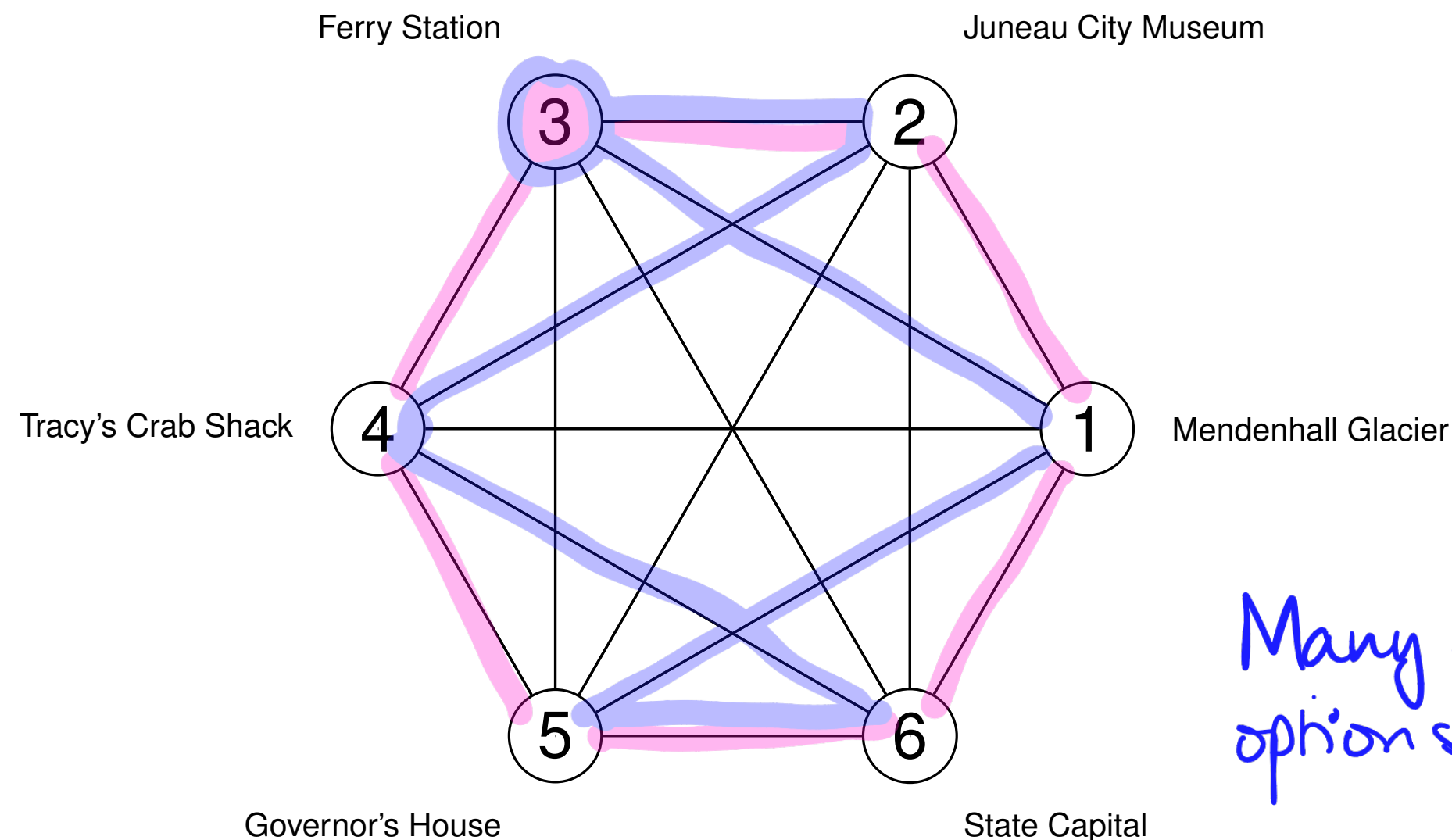
No EC or EP



No EC or EP

There's an easy way to tell if there's an Euler circuit:
Count how many odd degree vertices!

What would a Hamiltonian circuit in the graph below represent?



Many different options!

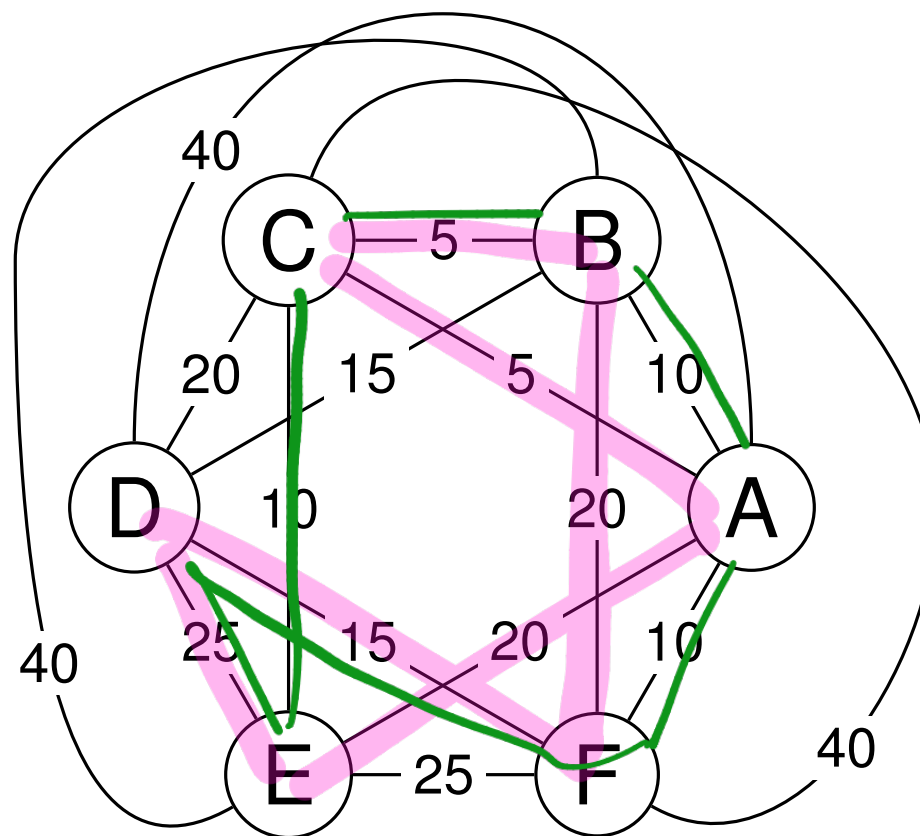
Why might you want a Hamiltonian circuit of smallest weight?

How might you find one??

Smallest weight might represent shortest distance!

How can we find the shortest hamiltonian circuit?

Computationally Hard Question!



$$\begin{aligned}\text{Total weight} &= \\ &5 + 5 + 20 + 25 + 15 + 20 \\ &= 90\end{aligned}$$

$$\begin{aligned}\text{Weight} &= 5 + 10 + 10 + 15 \\ &\quad + 25 + 10 \\ &= 75\end{aligned}$$

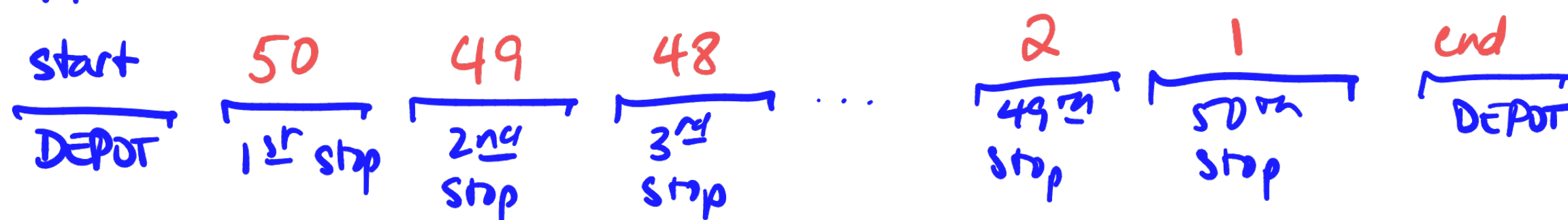
Comments on a concrete application and counting.

UPS/FedEx driver

Typically has 50-120 stops/day, start & end at distribution center. * Suppose UPS could find a shorter route?

When is an optimal eulerization easy? Under what conditions do you think it is easy to obtain an optimal eulerization?

Suppose 50 stops.



$$\# \text{ routes} = 50 \cdot 49 \cdot 48 \cdot \dots \cdot 3 \cdot 2 \cdot 1 = 50! > 3 \times 10^{64}$$

64 digits

"fifty factorial"

How big is 3×10^{64} ?

→ universe is ≈ 13.8 billion years.

billion = 10^9

million = 10^6

thousand = 10^3

trillion = 10^{12}

quadrillion = 10^{15}

$$(13.8 \times 10^9 \text{ years}) \left(\frac{365 \text{ days}}{\text{year}} \right) \left(\frac{24 \text{ hr}}{\text{day}} \right) \left(\frac{60 \text{ min}}{\text{hr}} \right) \left(\frac{60 \text{ s}}{\text{min}} \right)$$

$$\approx 4.3 \times 10^{17}$$

Suppose a computer could check 10^{16} routes/second.

Then it would take 3×10^{48} seconds. More than 30 times the age of the universe.